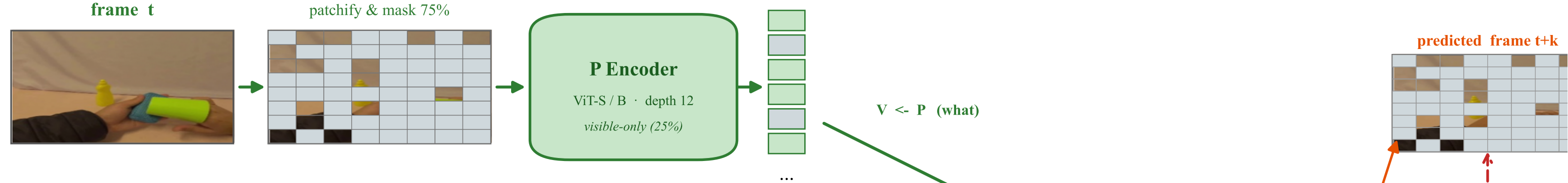


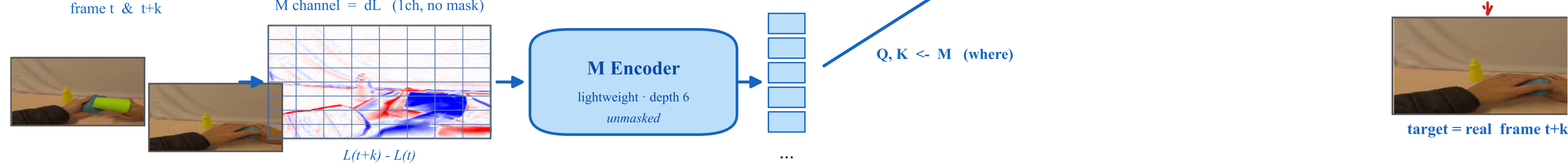
MCP-MAE — Motion-Conditioned Predictive MAE

Factorize appearance (P) and motion (M); M routes 'where' (Q,K) onto P's 'what' (V) to predict future pixels

P stream — appearance (WHAT)



M stream — motion (WHERE)



Unified mechanism — one predict() does both

M = real dL(t, t+k) -> predict future pixels (shown)
M = null dL(t, t)=0 -> identity routing -> reconstruct (gap=0)
 $L = \lambda * (L_t + L_{tk}) + \lambda * L_{pred}$ (motion size = gap)

vs SiamMAE — what changes

SiamMAE : two RGB frames, siamese enc · Q<-future / K,V<-past
MCP-MAE : factorize appearance P(RGB) & motion M(dL)
-> Q,K <- M (where) · V <- P (what) = what/where factorization
-> target = real pixels (no EMA / no JEPA -> collapse impossible)